

Generalization

INFO 521 · Module 2 · Lecture B — Bias-Variance & Regularization

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Learning outcomes

By the end of Lecture B you should be able to:

1. **Explain the bias-variance decomposition** of expected test error.
2. **State the tradeoff** and connect it to the U-curve.
3. **Derive and apply ridge regression**, and **tune λ by CV**.
4. **Connect ridge to the Module 1 uniqueness condition**.

Recap: the U, and why

Lecture A: test error is **U-shaped** in complexity — underfit on the left, overfit on the right.

Cross-validation told us *where* the bottom is. **Now: why is there a U at all?**

Bias and variance, intuitively

Imagine refitting on **many datasets** drawn the same way:

- **Bias** — the *systematic* miss: even on average, a too-simple model can't reach the truth.
- **Variance** — the *sensitivity*: a too-flexible model swings wildly from one dataset to the next.

Bias and variance on the data

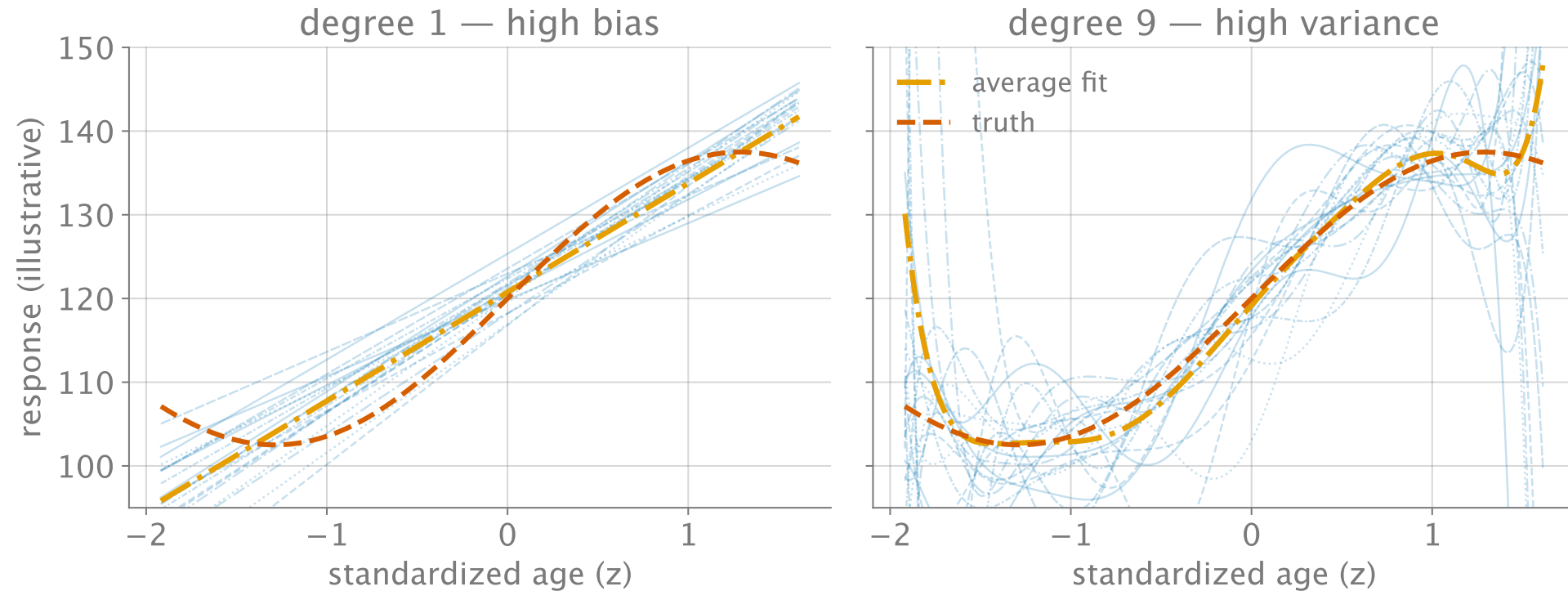


Figure 1: Repeated fits: degree 1 (biased, tight) vs. degree 9 (low-bias, high-variance).

The decomposition

For a fixed input $\mathbf{x}$, expected test error splits into three pieces:

$$\mathbb{E}[(y - \hat{f}(\mathbf{x}))^2] = \underbrace{\text{bias}^2}_{\text{systematic}} + \underbrace{\text{variance}}_{\text{sensitivity}} + \underbrace{\sigma^2}_{\text{irreducible}}.$$

The σ^2 floor is noise no model can beat. Full derivation → [appendix-bias-variance.html](https://www.stat.columbia.edu/gelman/appendix-bias-variance.html).

The tradeoff

As complexity rises:

- **bias² falls** — the model can represent more;
- **variance rises** — the model chases the sample;
- their **sum is the U**, resting on the irreducible σ^2 floor.

The best model is not the most flexible one — it is the one that **balances** the two.

Controlling the tradeoff: regularization.

Ridge regression

Add a penalty on the size of the weights:

$$\mathcal{L}_{\text{ridge}}(\mathbf{w}) = \|\mathbf{y} - \mathbf{X}\mathbf{w}\|_2^2 + \lambda \|\mathbf{w}\|_2^2.$$

- $\lambda = 0 \rightarrow$ ordinary least squares.
- $\lambda > 0 \rightarrow$ shrinks weights toward **0**, **trading a little bias for less variance**.

Ridge: the closed form

The penalty just adds $\lambda \mathbf{I}$ inside the normal equations:

$$\hat{\mathbf{w}}_{\text{ridge}} = (\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^\top \mathbf{y}$$

$\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I}$ is **positive definite for any** $\lambda > 0$ — so it is *always* invertible. Ridge **guarantees** the unique solution that Module 1 needed full column rank for.

The coefficient path

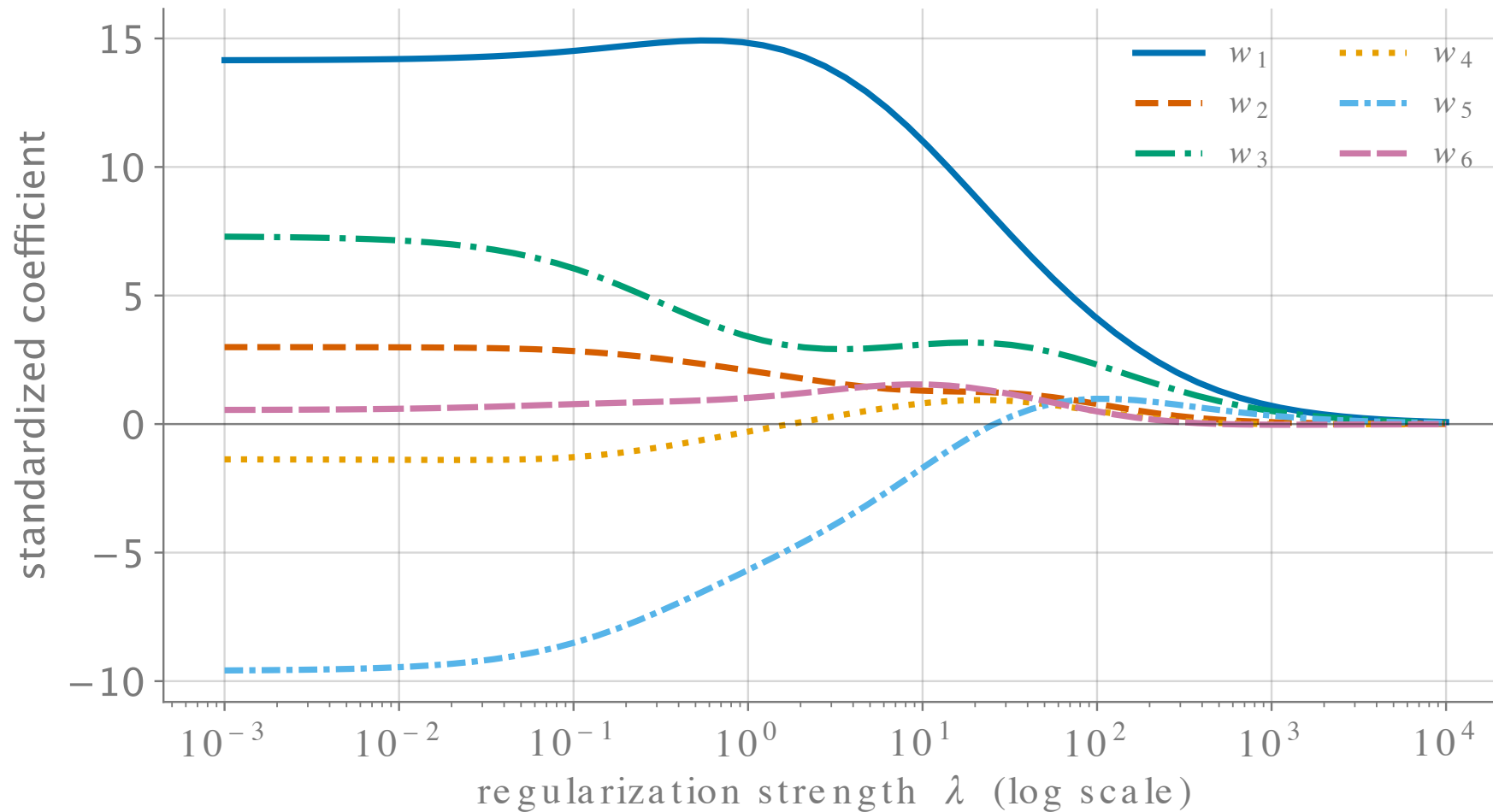


Figure 2: Ridge shrinks every coefficient toward 0 as λ grows (log- λ axis).

The two limits

- $\lambda \rightarrow 0$: the penalty vanishes — **ridge recovers OLS** ($\hat{\mathbf{w}}$ from Module 1). Maximum variance, minimum bias.
- $\lambda \rightarrow \infty$: the penalty dominates — $\hat{\mathbf{w}} \rightarrow \mathbf{0}$, the model predicts a constant. Maximum bias, minimum variance.

The useful λ lives in between — found by cross-validation.

Choosing λ by cross-validation

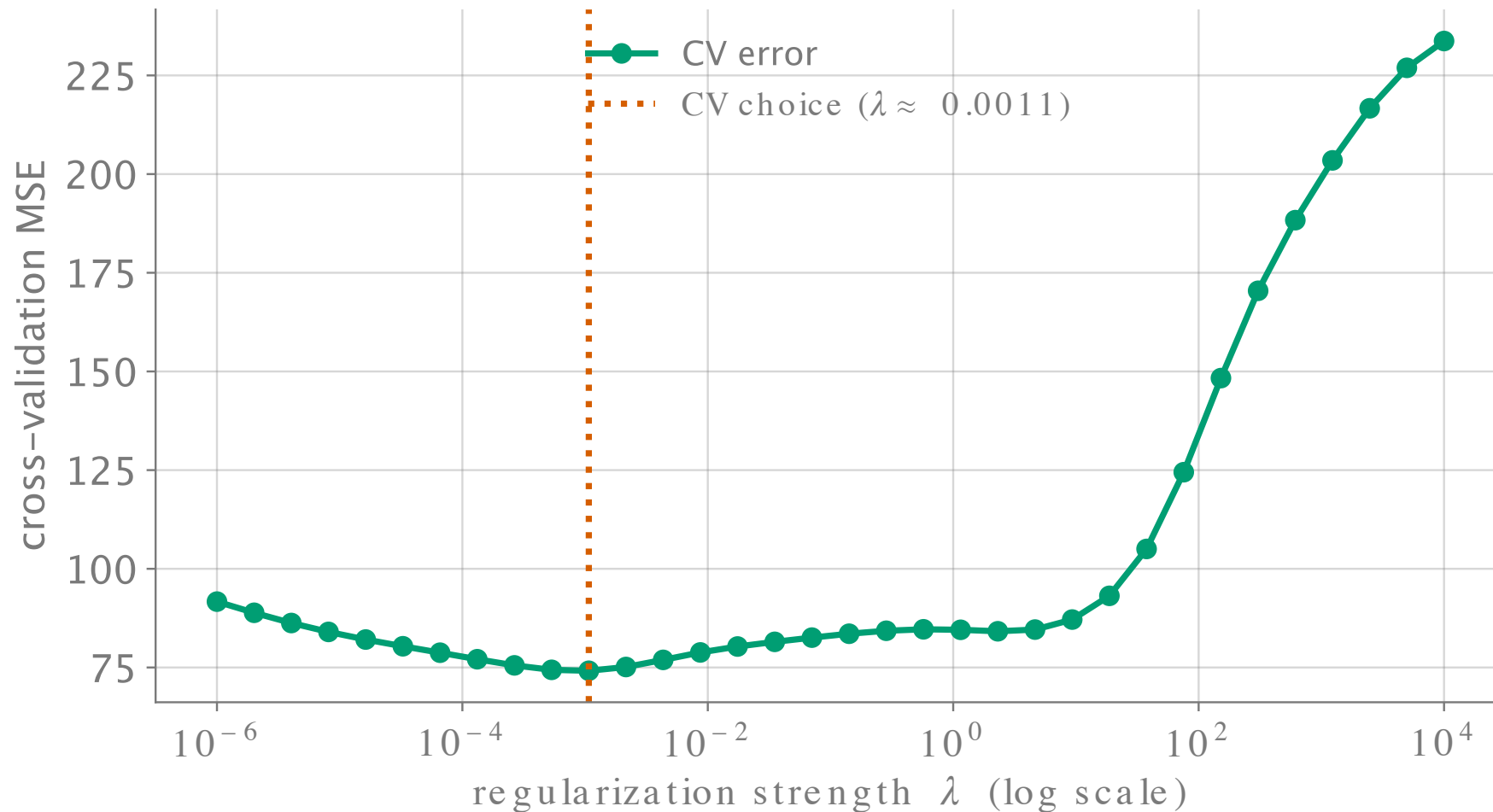


Figure 3: 5-fold CV error vs. λ for a degree-14 ridge fit; the minimum is the CV choice.

Live demo: bias-variance / ridge explorer

Bias, variance, and the cost of complexity

Each point is a patient's response at a given treatment dose; the true dose–response curve peaks at a moderate dose. Raise the polynomial degree to fit it and watch training vs. test error diverge, then use "Show 20 fits" to see variance directly and the ridge penalty λ to tame it.

$$\mathbb{E}[\big(y - \hat{f}(\mathbf{x})\big)^2] = \underbrace{\mathbb{E}[\hat{f}(\mathbf{x}) - f(\mathbf{x})]^2}_{\text{bias}^2} + \underbrace{\text{Var}[\hat{f}(\mathbf{x})]}_{\text{variance}} + \sigma^2$$

ridge regression trades variance for bias

$$\hat{\mathbf{w}} = \arg\min_{\mathbf{w}} \|\mathbf{y} - \mathbf{X}\mathbf{w}\|^2 + \lambda \|\mathbf{w}\|^2$$

(the intercept w_0 is left unpenalized, as is standard)

True curve Training points Your fit 20-fit overlay Test points

Polynomial degree

Ridge penalty λ

Degree

—

Ridge λ

—

Train MSE

Explorer — static fallback

The iframe above does not print or export to PDF; this snapshot stands in.

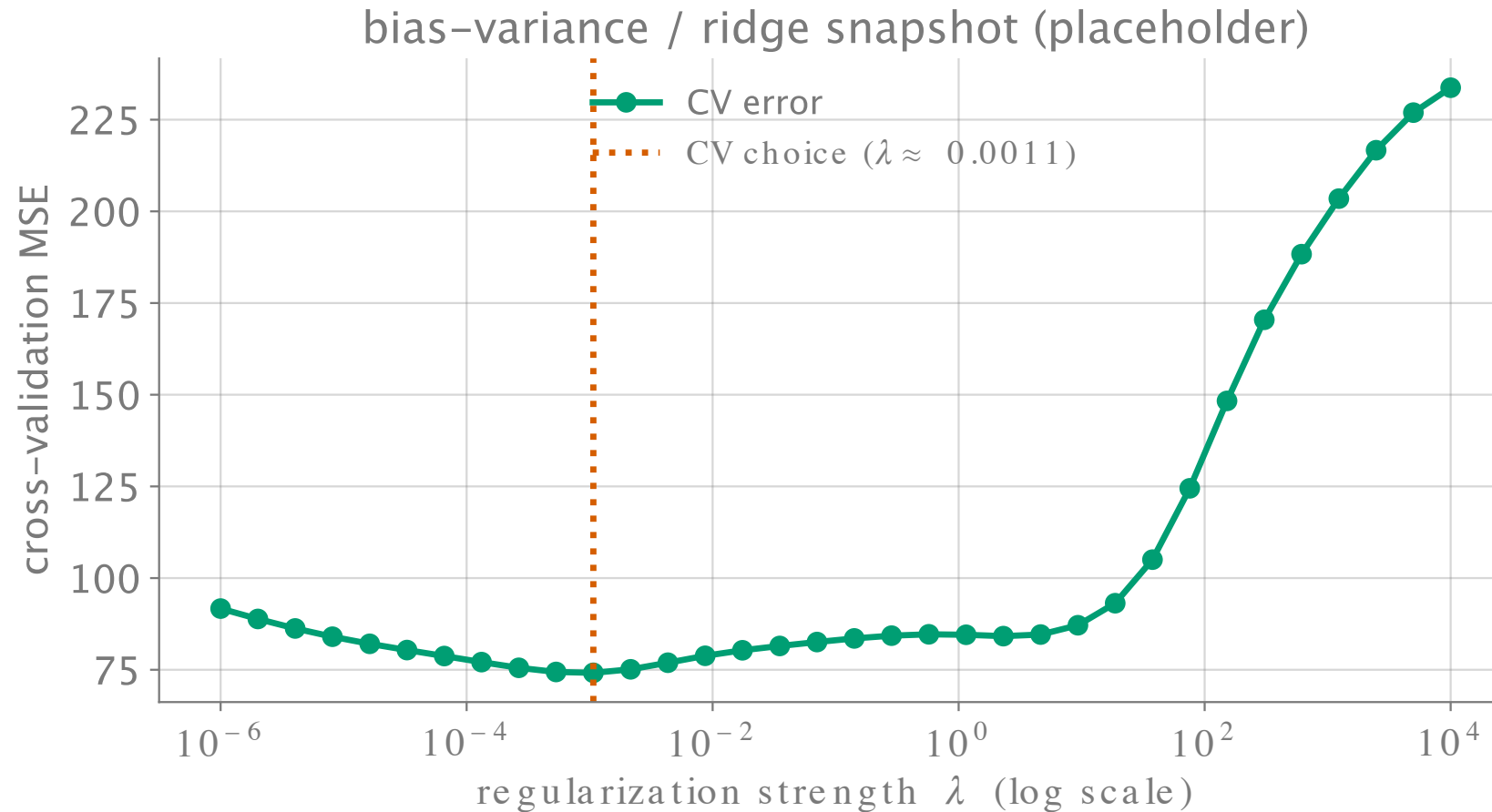


Figure 4: Static fallback: CV error vs. λ (used in print / PDF).

Forward hook

The ridge penalty $\lambda \|\mathbf{w}\|_2^2$ is secretly a **Gaussian prior** on $\mathbf{w}$: ridge is **MAP estimation** under that prior. We derive that in **Module 4**.

So “shrink the weights” and “believe, a priori, the weights are small” are the same statement — one more object doing double duty across the term.

Recap

- **Bias-variance** — test error = $\text{bias}^2 + \text{variance} + \sigma^2$; the U is their sum.
- **Ridge** — $\mathcal{L} + \lambda \|\mathbf{w}\|^2$, closed form $\hat{\mathbf{w}}_{\text{ridge}} = (\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^\top \mathbf{y}$; always invertible.
- **Tune λ by CV** — the knob that balances the tradeoff.

Module 2 has no standalone checkpoint; generalization discipline is assessed through your modeling choices in **Project 1**.